The Free Rational Calogero–Moser Flow as a Hermitian Pencil:
Global Scattering, Ordering Reversal, and the Route-A Boundary

Route-A structural certificate C196

Abstract

For every particle number $N \geq 2$, repulsive coupling $g > 0$, ordered real initial positions, and real momenta, we recover the rational Calogero–Moser trajectory as the ordered spectrum of one Hermitian pencil $Q_0 + tL_0$. A rank-one commutator proves that this pencil is simple for every real physical time. Eigenvalue perturbation then gives the Hamilton equations with the exact force coefficient $2g^2$, hence collision-free global completeness, while unitary transport of L_0 gives every trace integral and $\operatorname{Tr} L^2 = 2H$. Diagonalizing L_0 also gives a global scattering atlas: ordered asymptotic velocities and arbitrary real intercepts reconstruct one and only one ordered phase point. The two time ends reverse particle rank but preserve the intercept attached to each spectral line; distinct velocities also rule out bounded nonconstant periodic motion. These are classical Calogero–Moser/Moser structures assembled with explicit convention and scope locks, not a new priority claim.

Keywords: Calogero–Moser system; Hermitian pencil; Lax pair; complete flow; isospectral integrals.

中文摘要

对任意粒子数 $N \geq 2$、排斥耦合 $g > 0$、严格有序的实初始位置和实动量，本文把有理 Calogero–Moser 轨道写成单个厄米矩阵束 $Q_0 + tL_0$ 的有序谱。 秩一交换子证明该谱在全部物理时间内保持单重；特征值微扰给出系数恰为 $2g^2$ 的哈密顿 方程，从而同时得到无碰撞与全局完备性。L_0 的酉共轭还给出全部迹积分以及 $\operatorname{Tr} L^2 = 2H$。对角化 L_0 还得到一个全局谱坐标图：有序速度与任意实截距 唯一重建有序相点。两个时间端反转粒子顺序，却保持每条谱直线的截距；互异的渐近速度还排除 有界非恒定周期运动。这些结论属于经典理论的约定锁定与整体闭合，并不主张经典定理优先权。

关键词：Calogero–Moser系统；厄米矩阵束；Lax对；全局完备性。

1 Frozen Hamiltonian and matrix convention

On the chamber $q_1 < \cdots < q_N$, set

$$H(q,p) = \frac{1}{2}\sum_{j=1}^{N} p_j^2 + \sum_{1\leq j<k\leq N} \frac{g^2}{(q_j - q_k)^2}, \qquad g > 0. \tag{1}$$

With $Q_0 = \operatorname{diag}(q_1, \ldots, q_N)$, define the Hermitian matrix

$$(L_0)_{jk} = p_j\delta_{jk} + \frac{ig(1-\delta_{jk})}{q_j - q_k}, \qquad X(t) = Q_0 + tL_0. \tag{2}$$

The sign, unordered-pair normalization in (1), and physical clock t are part of the claim. Direct multiplication gives

$$[Q_0, L_0] = ig(J - I), \qquad J = ee^*, \quad e = (1, \ldots, 1)^T, \tag{3}$$

and

$$\operatorname{Tr} L_0^2 = \sum_j p_j^2 + 2\sum_{j<k} \frac{g^2}{(q_j - q_k)^2} = 2H. \tag{4}$$

Calogero's inverse-square model and Moser's isospectral solution retain classical ownership [1, 2].

2 A rank-one proof of simplicity

For all real t, $[X(t), L_0] = [Q_0, L_0] = ig(J - I)$. Suppose an eigenspace E of $X(t)$ had dimension at least two, and let P_E be its orthogonal projector. Because $X(t)|_E$ is scalar,

$$P_E[X(t), L_0]P_E = 0.$$

Equation (3) would force $P_E J P_E = P_E$. The left side has rank at most one, whereas the right side has rank at least two, a contradiction. Thus the increasingly ordered eigenvalues

$$x_1(t) < \cdots < x_N(t)$$

of $X(t)$ exist without a crossing for every $t \in \mathbb{R}$. The same argument, compressing (3) to an eigenspace of L_0, also proves that L_0 has simple spectrum.

3 Newton equations and global completeness

Choose a smooth unitary eigenbasis $U(t)$ of $X(t)$ and gauge its columns so that $e^*U(t) = (1, \dots, 1)$. The gauge is possible: the diagonal compression of (3) says that every overlap has modulus one. Put $\widetilde{L} = U^* L_0 U$. The off-diagonal part of the commutator gives

$$(x_j - x_k)\widetilde{L}_{jk} = ig \quad (j \neq k). \tag{5}$$

Hermitian eigenvalue perturbation gives

$$\dot{x}_j = \widetilde{L}_{jj}, \qquad \ddot{x}_j = 2\sum_{k\neq j} \frac{|\widetilde{L}_{kj}|^2}{x_j - x_k} = 2g^2 \sum_{k\neq j}(x_j - x_k)^{-3}.$$

At $t = 0$, $(x_j, \dot{x}_j) = (q_j, p_j)$, so these are precisely Hamilton's equations for (1). Local uniqueness on the chamber identifies the pencil curve with the physical solution. Since $X(t)$ exists and remains simple for every real t, the solution is collision-free and complete in both time directions.

Finally $L(t) = U(t)^* L_0 U(t)$, hence for every integer $m \geq 1$,

$$\operatorname{Tr} L(t)^m = \operatorname{Tr} L_0^m.$$

In particular (4) is the Hamiltonian integral.

4 A global spectral atlas

Let $\lambda_1 < \cdots < \lambda_N$ be the simple eigenvalues of L_0, and let v_a be normalized eigenvectors. Compressing (3) to $\mathbb{C}v_a$ shows $|e^* v_a| = 1$; fix the remaining phase by $e^* v_a = 1$. Define the real intercept coordinates

$$a_a = v_a^* Q_0 v_a.$$

In this gauged L_0-eigenbasis, the commutator determines every entry of $\widetilde{Q}$:

$$\widetilde{L} = \operatorname{diag}(\lambda_1, \dots, \lambda_N), \qquad \widetilde{Q}_{aa} = a_a, \quad \widetilde{Q}_{ab} = \frac{ig}{\lambda_b - \lambda_a} \quad (a \neq b). \tag{6}$$

Thus a phase point determines a unique pair $(\lambda_1 < \cdots < \lambda_N, a \in \mathbb{R}^N)$.

Conversely, start with any such pair and define the Hermitian matrix $\widetilde{Q}$ by (6). It obeys $[\widetilde{Q}, \widetilde{L}] = ig(J - I)$. The same compression/rank argument proves that $\widetilde{Q}$ has simple spectrum. Order its eigenvalues and diagonalize it. The diagonal compression of the commutator makes every component of the transformed e have modulus one; a diagonal phase gauge makes all components equal to one. In that basis

$$Q = \operatorname{diag}(q_1 < \cdots < q_N), \qquad L_{jk} = p_j \delta_{jk} + \frac{ig(1 - \delta_{jk})}{q_j - q_k},$$

with real p_j. This construction is inverse to the forward one, so (λ, a) is a global bijective spectral atlas, not merely asymptotic notation. No symplectic-form claim is needed here.

5 Both scattering ends and ordering reversal

Apply nondegenerate Hermitian perturbation theory to $t(L_0 + t^{-1}Q_0)$ as $t \to +\infty$. The gauge above identifies the first correction with a_j, giving

$$x_j(t) = t\lambda_j + a_j + O(t^{-1}), \qquad \dot{x}_j(t) = \lambda_j + O(t^{-2}). \tag{7}$$

For $t \to -\infty$, multiplication by the negative scalar reverses the order of the eigenvalues. Therefore

$$x_j(t) = t\lambda_{N+1-j} + a_{N+1-j} + O(|t|^{-1}), \qquad \dot{x}_j(t) = \lambda_{N+1-j} + O(t^{-2}). \tag{8}$$

The spectral line (λ_m, a_m) consequently has incoming ordered rank $N + 1 - m$ and outgoing ordered rank m. Its intercept is the same at both ends; "zero shift" here refers only to these source-native spectral-line coordinates, not to a target normalization.

Since L_0 is simple and $N \geq 2$, $\lambda_N - \lambda_1 > 0$. Equations (7)–(8) make the relative diameter grow linearly in both time directions. Hence every orbit is unbounded in relative configuration and there is no bounded nonconstant periodic orbit.

6 Executable certificate and strict Route-A stop

The deterministic oracle uses 18 rational systems ($2 \leq N \leq 7$, three couplings), 126 pencil times, 417 exact Hermitian and 417 exact commutator entry checks, and 99 exact trace/energy checks. Its sampled gap is at least 0.7215; the largest Newton, atlas-matrix, and inverse-position residuals are below 4.5×10^{-15}, 6.5×10^{-15}, and 6.3×10^{-15}. A producer-independent realified-Jacobi/projector checker passes 2,210 assertions, a separate SymPy reconstruction passes 1,200 checks, replay is byte exact, and 135 repaired-hash plus one stale-hash attacks are rejected. The repaired attacks include unknown-key injections at every finite schema level. These finite results test signs and implementations; the all-N quantifier belongs to the proof and classical source theorem.

Continuous particle coordinates and g have no intrinsic rational-prime origin. The complete free flow has no periodic-orbit carrier, source zeta, target divisor, functional equation, or Weil-type compression. The source-native quantum expression

$$-\frac{1}{2}\Delta + \sum_{j<k} \frac{g^2}{(q_j - q_k)^2}$$

on the ordered chamber has a natural semibounded Friedrichs realization, but no target spectral identification is made. Thus the frozen verdict is

$$\begin{aligned}(A0, A1, A2, A3, A4) = (&\mathrm{A0\_FAIL}, \mathrm{A1\_FAIL}, \mathrm{A2\_FAIL},\\ &\mathrm{A3\_FAIL}, \mathrm{A4\_NATURAL\_QUANTIZATION}),\end{aligned}$$

overall `ROUTE_A_REJECTED`, Route B false, under `NO_BAD_EULER_OR_ROOT_NUMBER`.

Degenerate and model boundaries. At $g = 0$ free trajectories may cross; coincident initial positions are singular phase points, and $N = 1$ is a trivial free particle. Confining, trigonometric, hyperbolic, elliptic, spin, complex, and quantum spectral variants are outside the theorem. No finite Artin–Mazur zeta, prime-orbit law, Hilbert–Polya operator, external review, or classical priority is claimed.

Round-2 / final increment. The final revision adds both asymptotic ends with rank reversal and preserved spectral-line intercepts, proves the absence of bounded nonconstant periodic motion, exposes exact validation totals, and closes the natural-quantization boundary with the strict five-gate Route-A verdict.

Declarations

Data and code availability. Package-local deterministic code and exact evidence accompany this manuscript.

Ethics. No human, animal, clinical, personal, or sensitive data are used; approval is not applicable.

Funding and conflicts. No external funding is reported; no conflict of interest is known.

Author contributions (CRediT). This anonymous certificate records Conceptualization, Formal analysis, Software, Validation, Writing—original draft, and Writing—review and editing. AI systems are not authors.

AI-use disclosure. An AI coding assistant supported drafting and exact-code development; it was not an external reviewer or independent error process.

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